

Introduction to K-theory and Hochschild Homology

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Abstract

We introduce low dimensional K-theory first, and briefly explain Quillen's plus construction. Then we recall the Hochschild homology of an algebra, and its relation with K-theory. If time permits, we give a briefly introduction on stable homotopy theory, and v_n -periodicity.

1 What is K-theory

Within the study of algebra, (unital) rings are fundamental objects, and (finite rank) vector bundles over a space can be seen as an analogue of finitely generated projective modules over a ring. A natural question is how to classify finitely generated projective modules over a ring.

A naive idea is to take the set of isomorphism classes of f.g. proj. modules, and try to analysis this set. We use $P(R)$ to denote the set, one observes that $P(R)$ can be equipped with a commutative monoid structure, and the operation is given by direct sum.

Definition 1.1. The group completion of an abelian monoid M is a group M^{gp} , together with a monoid map $M \rightarrow M^{gp}$ which is universal in the sense that, for any abelian group A and every monoid map $\alpha : M \rightarrow A$, there is a unique abelian group homomorphism $M^{gp} \rightarrow A$, such that a diagram commutes.

It showed that for any monoid the group completion of it exists.

Proposition 1.2. For any abelian monoid M , let $M^{gp} = F(M)/R(M)$. Here $F(M)$ is the free abelian group generated by M , and $R(M)$ is the subgroup of $F(M)$ generated by $[m+n] - [m] - [n]$. Then M^{gp} is the group completion of M up to isomorphism.

Definition 1.3. Let R be an unital ring, then the 0th K-group is defined to be $K_0(R) := (P(R))^{gp}$.

In fact, the 0th K-group is a commutative ring by the fact that $P(R)$ can be equipped with a commutative semi-ring structure.

Example 1.4. 1. For any field F , we have $K_0(F) \cong \mathbb{Z}$. This is follows, because any finitely generated module over F is free.

2. Let R be a commutative ring, then we have following equivalent statement.

$$K_0(R) \cong \mathbb{Z} \iff \text{every finitely generated } R\text{-module is stably free.}$$

The ($\Leftarrow$) is relative easy. Because if $M \in P(R)$, then $[M \oplus R^k] = [R^m] = [M] + [R^k]$. This implies that $[M] = (m-k)[R]$, so $K_0(R)$ is generated by $[R]$.

3. Let R be a Dedekind domain, i.e. R is Noetherian integrally closed domain such that every non-zero prime ideal is maximal. Then

$$K_0(R) \cong \mathbb{Z} \oplus \text{Pic}(R).$$

And the isomorphism is determined by rank map and determinant map.

For any associative ring R , let $[R, R]$ denote the subgroup of R generated by elements $rs - sr$, $r, s \in R$. For each n , the trace of an $n \times n$ matrix provides an additive map from $M_n(R)$ to $R/[R, R]$. Note that $\text{tr}(AB) = \text{tr}(BA)$. If P is a finitely generated projective R -module, then there is an isomorphism $P \oplus Q \cong R^n$, for some n , and Q . Thus, there is an idempotent $e \in M_n(R)$, such that $P = eM_n(R)$. So $\text{End}(P) = eM_n(R)e$, now $\text{tr}(\text{id}_P) \in R/[R, R]$. It can be showed that, this trace is independent of choice of Q and R^n , and e . Furthermore, it can be showed that this yields an group homomorphism

$$K_0(R) \rightarrow R/[R, R].$$

Let $\text{GL}_n(R)$ be the group of invertible matrices of size n , let $E_n(R)$ be the subgroup of $\text{GL}_n(R)$ generated by elementary matrices $e_{i,j}(\lambda)$, $1 \leq i \neq j \leq n$, $\lambda \in R$. Let $\text{GL}(R) = \bigcup_{n \geq 1} \text{GL}_n(R)$, where $\text{GL}_n(R)$ viewed as a subgroup of $\text{GL}_{n+1}(R)$ via natural embedding. Similarly, $E(R) = \bigcup_{n \geq 1} E_n(R)$.

Definition 1.5. The first K-group is defined to be $K_1(R) := \text{GL}(R)/E(R)$. It can be showed that $E(R) = [\text{GL}(R), \text{GL}(R)]$, so $K_1(R) = \text{GL}(R)^{\text{ab}} \cong H_1(\text{GL}(R), \mathbb{Z})$.

Example 1.6. Let R be a commutative ring. The determinant gives a surjection $\text{GL}(R) \rightarrow R^\times = \text{GL}_1(R)$ split by injection $\text{GL}_1(R) \rightarrow \text{GL}(R)$. Then we have a short exact sequence

$$0 \rightarrow SK_1(R) \rightarrow K_1(R) \rightarrow R^\times \rightarrow 0.$$

If R is the ring of integers in an algebraic number field, then $SK_1(R) = 0$.

Definition 1.7. The n th Steinberg group $\text{St}_n(R)$ is defined to the quotient $F(X)/Q$, where $X = \{x_{i,j}^{(n)}(\lambda) | 1 \leq i \neq j \leq n, \lambda \in R\}$. Here, Q is the subgroup generated by

1. $x_{i,j}^{(n)}(\lambda)x_{i,j}^{(n)}(\mu)x_{i,j}^{(n)}(\lambda + \mu)^{-1}$, for all i, j , and all $\lambda, \mu \in R$.
2. $[x_{i,j}^{(n)}(\lambda), x_{k,l}^{(n)}(\mu)]$, for $i \neq l, j \neq k$, for all $\lambda, \mu \in R$.
3. $[x_{i,j}^{(n)}(\lambda), x_{j,k}^{(n)}(\mu)]x_{i,k}^{(n)}(\lambda\mu)^{-1}$, for $i \neq k$, for all $\lambda, \mu \in R$.

Therefore there is a natural map $\text{St}(R) \rightarrow E(R)$, and the second K-group is defined to be the kernel of this map.

Example 1.8. 1. It can be showed that $K_2(R) \cong H_2(E(R), \mathbb{Z})$.

2. Let F be a field, then $K_2(F) \cong K_2^M(F)$, where $K^M(-)$ is the Milnor K-theory.

Definition 1.9. Let F be a field. Consider the tensor algebra of the group ($\mathbb{Z}$ -module) $F^\times$,

$$T(F^\times) := \bigoplus_{m \geq 0} (F^\times)^{\otimes m} = \mathbb{Z} \oplus F^\times \oplus (F^\times \otimes F^\times) \oplus \dots$$

Let Q be the ideal of $T(F)$ generated by $x \otimes (1 - x)$, where $x \in F \setminus \{0, 1\}$. Then $K_*^M(F) := T(F^\times)/Q$. So, $K_2^M(F)$ is the quotient of free abelian group generated by symbols $x \otimes y$, for $x, y \in F^\times$, modulo the following relations.

1. $xx' \otimes y = (x \otimes y)(x' \otimes y), x \otimes yy' = (x \otimes y)(x \otimes y')$;
2. $x \otimes (1 - x) = 1$, for $x \in F \setminus \{0, 1\}$.

There is map (norm-residue) from Milnor K-theory to étale cohomology.

1. Abstractly, this map is induced by cup product of étale cohomology.

$$F^\times \otimes F^\times \rightarrow H^1(F, \mu_n) \otimes H^1(F, \mu_n) \rightarrow H^2(F, \mu_n^{\otimes 2}).$$

It can be showed that $a \otimes (1 - a)$ maps to 0, hence the map above yields the following map

$$K_2^M(F) \rightarrow H^2(F, \mu_n^{\otimes 2}) = H_{\text{ét}}^2(F, \mathbb{Z}/n(2)).$$

2. More concretely, take $(a, b) \in F^\times \times F^\times$. If the field F contains a primitive n th root of unity, ζ . Then the cyclic algebra $A = A_\zeta(a, b)$ is defined to be the quotient $F\langle x, y \rangle / Q$. Here, Q is generated by $x^n - a, y^n - b, yx = \zeta xy$. It's not hard to show that $\{x^i y^j | 1 \leq i, j < n\}$ forms a basis of A . And it's not hard to show that A is central simple algebra over F . Moreover, $A^{\otimes n}$ is a matrix algebra. Thus, this construction yields a map

$$F^\times \otimes F^\times \rightarrow \text{Br}(F).$$

And $A_\zeta(a, 1 - a)$ is equivalent to matrix algebra, this follows from the fact that $(x + y)^n = 1$ in A . So, there is a map

$$K_2(F) \rightarrow \text{Br}(F).$$

Recall the Kummer sequence

$$1 \rightarrow \mu_n \rightarrow \mathbb{G}_m \rightarrow \mathbb{G}_m \rightarrow 1.$$

Applying $H_{\text{ét}}^*(F, -)$, we have $H^1(F, \mu_n) \simeq F^\times / nF^\times$, and $H^2(F, \mu_n) \simeq \text{Br}(F)[n]$.

Proposition 1.10. *The Kummer isomorphism extends uniquely to a graded ring homomorphism*

$$h_F : K_*^M(F)/n \rightarrow H^*(F, \mu_n^{\otimes *}).$$

The individual maps $K_i^M(F)/n \rightarrow H^i(F, \mu_n^{\otimes i})$ are called the norm residue symbols, and also higher Galois symbols.

Theorem 1.11 ([HW19, Theorem A]). *If $n \in F^\times$, then the norm residue maps are isomorphisms.*

The above theorem is a special case of the following.

Theorem 1.12. *Let X be a smooth variety over a field containing $1/\ell$. Then the map*

$$H_{\text{mot}}^{p,q}(X, \mathbb{Z}/\ell) \rightarrow H_{\text{ét}}^p(X, \mu_\ell^{\otimes q}) = H_{\text{ét}}^p(X, \mathbb{Z}/\ell(q)).$$

This would imply the Lichtenbaum–Quillen conjecture

Conjecture 1.13. *Let $\mathcal{O}$ be a nice ring. There is a spectral sequence*

$$E_2^{*,*} = H_{\text{ét}}^*(\mathcal{O}, \mathbb{Z}/\ell(*)) \implies K_*(\mathcal{O}, \mathbb{Z}/\ell)$$

that converges to the target at large degrees.

In fact, there is a motivic spectral sequence

$$E_2^{p,q} = H_{\text{mot}}^{p-q}(X, \mathbb{Z}/\ell(-q)) \implies K_{-p-q}(X, \mathbb{Z}/\ell),$$

where X is a smooth scheme over a field k .

We sketch the construction of K-theory space.

Definition 1.14. Let R be an unital ring, the K-theory space of R is defined to be $K_0(R) \times B\text{GL}(R)^+$.

Recall the bar construction (classifying space) of a group G is defined to be the geometric realization of the nerve of category $*G$. Here, $*G$ is the category with one object, and the hom set is bijectively equivalent to G . The nerve of $*G$ is $N_\bullet(*G) = G^{\times \bullet}$.

The geometric realization of a simplicial set $X_\bullet$ is defined to be the quotient space

$$\bigsqcup_{n \geq 0} X_n \times \Delta^n / (x, f_*(t)) \sim (X(f)(x), t),$$

here $f \in \text{hom}([m], [n]), x \in X_n, t \in \Delta^m$. The plus construction of a space is defined by following property.

Definition 1.15. The map $B\text{GL}(R) \rightarrow B\text{GL}(R)^+$ is universal for maps into H -spaces. Concretely, the plus construction is characterized by following property.

1. There is a isomorphism $\pi_1(X^+) \simeq \pi_1(X)^{ab}$.
2. The map induce isomorphism on homology

$$H_n(X, \mathbb{Z}) \rightarrow H_n(X^+, \mathbb{Z}).$$

3. The above two properties is universal. In other words, for any Y and map $X \rightarrow Y$ satisfying above conditions. Then there is a unique map $X^+ \rightarrow Y$ such that following diagram commutes

$$\begin{array}{ccc} X & \longrightarrow & X^+ \\ & \searrow & \downarrow \\ & & Y \end{array}$$

Construction 1.16. We sketch the construction. First by attaching two cells, one could easy kill the commutators subgroup of the fundamental group of X , we denote the space by X_1 . But this may changes the homology of X . Since the relative cells of (X_1, X) has only 2-cells. By attaching 3-cells carefully will making the relative homology to be 0.

In fact, the K-theory space is a infinite loop space.

Definition 1.17. A space X is called 1-fold loop space, if there is a space X_1 such that $\Omega X_1 \cong X$. A space is called n -fold loop space, if there is an $n - 1$ -fold loop space X_{n-1} , such that $\Omega X_{n-1} \cong X_n$. A space X is called ∞ -loop space, if X is m -fold loop space, for any $m \in \mathbb{N}$.

Remark 1.18. The category of ∞ -loop space is in fact equivalent to the category of connective spectra $\text{Sp}_{\geq 0}$. Furthermore, $\text{Sp}_{\geq 0}$ is equivalent to the category of group like $\mathbb{E}_\infty$ -monoids, we denote the later category as $\text{CMon}(\text{Spc})^{gp}$. The topological commutative monoid $\bigsqcup_{n \geq 1} B \text{GL}_n(R)$ is an $\mathbb{E}_\infty$ -monoid. The natural forgetful functor

$$\text{CMon}(\text{Spc})^{gp} \rightarrow \text{CMon}(\text{Spc})$$

admits a left adjoint, which called group completion. In particular, the space $\Omega B(\bigsqcup_{n \geq 1} B \text{GL}_n(R)) \simeq B \text{GL}(R) \times \mathbb{Z}$ is the group completion of $\bigsqcup_{n \geq 1} B \text{GL}_n(R)$.

2 What is Hochschild homology

Let k be a commutative ring, A be a k -algebra. Then the Hochschild homology of A relative to k is defined as follows.

Definition 2.1. The Hochschild homology $\text{HH}(R/k)$ is defined to be the following chain complex.

$$\cdots \rightarrow R^{\otimes_k n} \rightarrow R^{\otimes_k (n-1)} \rightarrow \cdots R^{\otimes_k 2} \rightarrow R.$$

The differential is given by

$$d(x_0 \otimes \cdots \otimes x_n) = \sum_{i=0}^{n-1} (-1)^i (\cdots \otimes x_i x_{i+1} \otimes \cdots) + (-1)^n x_n x_0 \otimes \cdots \otimes x_{n-1}.$$

In other words, it is the Moore complex of a simplicial k -module. $\text{HH}(R/k)_n = R^{\otimes_k n+1}$, and the face map is given by $d_i(x_0 \otimes \cdots \otimes x_n) = (\cdots x_i x_{i+1} \otimes \cdots)$, $i < n$, $d_n(x_0 \otimes \cdots \otimes x_n) = (x_n x_0 \otimes \cdots \otimes x_{n-1})$. And for any associative ring R , the Hochschild homology $\text{HH}(R) := \text{HH}(R/\mathbb{Z})$.

Remark 2.2. The above definition in fact required that R to be a flat k -module. In the following, we define the Hochschild homology by

$$\text{HH}(R) := R \otimes_{R \otimes_k^{\mathbb{L}} R^{op}} R.$$

One sees that if R is flat over k , then the previous definition is a resolution of R as R -bimodule.

Example 2.3. 1. $\mathrm{HH}_0(R/k) = R/\mathrm{imd}$. Note that $d(x_0 \otimes x_1) = x_0 x_1 - x_1 x_0$, therefore $\mathrm{HH}_0(R/k) = R/[R, R]$.

2. Let R be a commutative ring. Then $\mathrm{HH}_0(R) = R$, and $\mathrm{HH}_1(R) = \ker(d_1)/\mathrm{imd}_2 \cong \Omega_{R/\mathbb{Z}}^1$. Note that $\ker d_1 = R^{\otimes 2}$, and we have an isomorphism given by

$$[x \otimes y] \mapsto xdy \in \Omega_{R/\mathbb{Z}}^1$$

with inverse $xdy \mapsto [x \otimes y]$.

3. A remainder of Kahler differentials. Let $k \rightarrow R$ be a map of commutative rings. Then $\Omega_{R/k}^1$ is defined to be the free R -module $\bigoplus_{r \in R} Rdr$ modulo following relations.

- (a) $d(r + s) = dr + ds$, for all $r, s \in R$.
- (b) $d(rs) = dr \cdot s + r \cdot ds$, for all $r, s \in R$.
- (c) $d(t) = 0$, for all $t \in k$.

4. Hence, as seen before. There is a trace map

$$K_0(R) \rightarrow \mathrm{HH}_0(R) \cong R/[R, R].$$

A natural question arises, is there a map of spectra

$$K(R) \rightarrow \mathrm{HH}(R),$$

such that after taking π_0 coincides with the classical trace map $K_0(R) \rightarrow R/[R, R] \cong \mathrm{HH}_0(R)$?

Roughly speaking, the Dennis trace map can be given by following maps.

$$N_\bullet(\mathrm{GL}_n(R)) \rightarrow N_\bullet^{cy}(\mathrm{GL}_n(R)) \rightarrow N_\bullet^{cy}(M_n(R)) \simeq N_\bullet^{cy}(R).$$

Here the first map is given by

$$\begin{aligned} N_n(\mathrm{GL}_n(R)) &\rightarrow N_n^{cy}(\mathrm{GL}_n(R)) \\ (g_1, \dots, g_n) &\mapsto ((\prod g_i)^{-1}, g_1, \dots, g_n) \end{aligned}$$

After suitable stabilization, we get a trace map

$$B\mathrm{GL}(R)^+ \rightarrow \mathrm{HH}(R).$$

Let's see some examples.

Example 2.4. 1. The Hochschild homology $\mathrm{HH}_*(\mathbb{F}_p/\mathbb{Z}) \cong \Gamma_{\mathbb{F}_p}(x)$, where $|x| = 2$.

2. $\mathrm{HH}(\mathbb{Q}[x_1, \dots, x_n]/\mathbb{Z}) \otimes_{\mathbb{Q}} \mathrm{HH}(\mathbb{Q}/\mathbb{Q}) \simeq \mathrm{HH}(\mathbb{Q}[x_1, \dots, x_n]/\mathbb{Q})$. This has homotopy group as

$$\mathrm{HH}_*(\mathbb{Q}[x_1, \dots, x_n]/\mathbb{Q}) \simeq \mathbb{Q}[x_1, \dots, x_n] \otimes \Lambda(dx_1, \dots, dx_n).$$

Remark 2.5. Since $\mathbb{F}_p$ can be resolved by $\mathbb{Z} \rightarrow \mathbb{Z} \rightarrow \mathbb{F}_p$, then $\mathbb{F}_p$ is equivalent to the CDGA $\mathbb{Z}[\alpha]/\alpha^2$, $d\alpha = p$, and $|\alpha| = 1$. Thus $\mathbb{F}_p \otimes^{\mathbb{L}} \mathbb{F}_p \simeq \Lambda_{\mathbb{F}_p}(\alpha)$. Now one checks

$$\Gamma_A(x) := A[x_1, x_2, \dots]/(x_i x_j = \binom{i+j}{i} x_{i+j}), \quad dx_i = \alpha x_{i-1}, \quad |x_i| = 2i$$

is a resolution of $\mathbb{F}_p$ as A -algebra. First, it is easy to verify that $\Gamma_A(x)$ is a resolution. We show that $x_1 x_n = (n+1)x_{n+1}$. We do the induction on n , for $n = 1$, we need to show $x_1^2 = 2x_2$. This follows from the fact that $d(x_1^2 - 2x_2) = 2\alpha x_1 - 2\alpha x_1 = 0$. Since degree 4 space is generated by γx_2 , where $\gamma \in \mathbb{F}_p$. Hence we know that $x_1^2 - 2x_2 = 0$. Suppose $x_1 x_{n-1} = nx_n$, then $d(x_1 x_n - (n+1)x_{n+1}) = x_1 \alpha x_{n-1} + \alpha x_n - (n+1)\alpha x_{n-1} = 0$. A same argument shows that $x_1 x_n = (n+1)x_{n+1}$.

The second example follows from Koszul resolution.

As seen before, the K-theory space lies naturally in the category of spectrum. And for Hochschild homology, there is also a spectral version. The spectral version of Hochschild homology has better behavior, for instance $\mathrm{THH}_*(\mathbb{F}_p) \simeq \mathbb{F}_p[u]$, $|u| = 2$.

3 Stable homotopy and v_n -periodicity

Recall that a generalized cohomology theory is a functor

$$Ho(\mathbf{Spc})^{op} \rightarrow \mathcal{A}b$$

satisfies some axioms.

Theorem 3.1. *Any generalized cohomology theory E^* is represented by a sequence of spaces $\{E_n\}_{n \geq 0}$ such that $E^k(X) \simeq [X, E_k]$ and the maps $E_k \rightarrow \Omega E_{k+1}$ are weak homotopy equivalence.*

Definition 3.2. An Ω -spectrum is a sequence of based space T_n and weak homotopy equivalences $\tilde{\sigma} : T_n \rightarrow \Omega T_{n+1}$.

The category of spectrum can be made to a presentable stable ∞ -category, and it can be also equipped with a symmetric monoidal structure. We denote the category by $\mathbf{Sp}$. This is our new based category, instead of category of abelian groups. The tensor unit in $\mathcal{A}b$ is $\mathbb{Z}$, while the unit in $\mathbf{Sp}$ is the sphere spectrum $\mathbb{S}$. For any ring R , there is a (ring) spectrum $HR \in \mathbf{Sp}$ corresponds to R . In the sense, that $\pi_*(HR) \simeq \pi_0(HR) \simeq R$. Let's see a more topological example.

Remark 3.3. An ∞ -category should be viewed as a category whose hom set equipped with topology, i.e. topological category. In an ∞ -category, the composition law is not strictly satisfied, but it's satisfied up to homotopy. And the (homotopy) reason makes the composition law valid, it's also unique up to homotopy, so on so forth... Let's see an example. For a space X , we define the ∞ -fundamental groupoid $\Pi_\infty(X)$ of X to be the category whose object are points of X . The 1-morphism is defined to be the path from x to y . Thus would make no sense to require a strict associative composition law. Say $\alpha_1 : x \rightarrow y, \alpha_2 : y \rightarrow z, \alpha_3 : z \rightarrow \cdot$. Then we only have $(\alpha_1 \circ \alpha_2) \circ \alpha_3 \simeq \alpha_1 \circ (\alpha_2 \circ \alpha_3)$, which is not a equality. The 2-morphism is defined to be homotopy between two paths, i.e. $h : \alpha \rightarrow \beta$, where $\alpha, \beta \in \text{hom}(x, y)$. Similarly, one can also define n -morphisms.

Definition 3.4. Let X be a space. Let M_1, M_2 be two closed m -dimensional manifolds and let

$$f_1 : M_1 \rightarrow X, \quad f_2 : M_2 \rightarrow X$$

be two continuous maps. These two maps are said to be bordant if there is a manifold W and a map $f : W \rightarrow X$ such that the boundary of W is $X_1 \sqcup X_2$, and the restriction of f on the boundary is $f_1 \sqcup f_2$. Then $\text{MU}_m(X) := \{f : M \rightarrow X\} / \sim$. Here, we require that M to be a stably complex manifold.

It would not be surprised that there is a spectrum MU , represents the complex cobordism theory. It is well known that MU has homotopy groups as

$$\text{MU}_* \simeq \mathbb{Z}[x_1, x_2, \dots], \quad |x_i| = 2i.$$

In fact, MU is an $\mathbb{E}_\infty$ -ring spectrum, and the above equivalence is a ring isomorphism. There are more ring spectrum constructed based on MU , for instance.

Example 3.5. 1. Using the Landweber exact functor theorem, one shows that $\mathbb{Z}_{(p)}[v_1, v_2, \dots]$ induced a homology theory. Hence, there is a spectrum BP such that $\text{BP}_* \cong \mathbb{Z}_{(p)}[v_1, v_2, \dots]$.

2. Similarly, there are spectra $\text{BP}\langle n \rangle$, such that $\text{BP}\langle n \rangle_* \simeq \mathbb{Z}_{(p)}[v_1, \dots, v_n]$.

3. The Morava K-theory $K(n)$, has the property $K(n)_* \simeq \mathbb{F}_p[v_n^\pm]$.

Remark 3.6. The Brown–Peterson can be equipped with an $\mathbb{E}_4$ -ring structure, and $\text{BP}\langle n \rangle$ can be equipped with an $\mathbb{E}_3$ -ring structure. The Morava K-theory can be equipped with an $\mathbb{E}_1$ -ring structure, moreover, there can not be an $\mathbb{E}_2$ -ring structure on $K(n)$. And there does not exists an $\mathbb{E}_{12}$ -ring structure on $\text{BP}, \text{BP}\langle n \rangle$.

Definition 3.7. Let X be a p -local finite spectrum, and $n \geq 0$. A self-map $v : \Sigma^k X \rightarrow X$ is said to be a v_n self-map if

$$K(m)_* v = \begin{cases} \text{multiplication by a rational number,} & \text{if } m = n = 0, \\ \text{isomorphism,} & \text{if } m = n \neq 0, \\ \text{nilpotent,} & \text{otherwise.} \end{cases}$$

Let $\mathcal{C}_0$ be the homotopy category of p -local finite spectra, we denote $\mathcal{C}_n, n \geq 1$ the full subcategory of $\mathcal{C}_0$ with $K(n-1)_*(X) = 0, \forall X \in \mathcal{C}_n$.

Theorem 3.8. *A p -local finite spectrum X admits a v_n self-map iff $X \in \mathcal{C}_n$. And if X admits a v_n self-map, then exists $N \in \mathbb{N}$, and $v : \Sigma^{2(p^n-1)p^N} X \rightarrow X$, such that*

$$K(m)_* v = \begin{cases} v_n^{p^N}, & \text{if } m = n, \\ 0, & \text{otherwise.} \end{cases} \quad (3.1)$$

For instance, it has been showed that for p odd, there is a v_1 -self map $\Sigma^{2p-2}\mathbb{S}/p \rightarrow \mathbb{S}/p$ for $\mathbb{S}/p$. For $p = 2$, there is a v_1 -self map $v_1^4 : \Sigma^8\mathbb{S}/2 \rightarrow \mathbb{S}/2$. The telescope spectrum is defined to be $T(n) := v_n^{-1}\mathbb{S}/(p, v_1, \dots, v_{n-1})$.

4 Trace method and cyclotomic boundedness

Definition 4.1. Let $A \in \text{Alg}(\text{Sp})$, we define the topological Hochschild homology to be $\text{THH}(A) := A \otimes_{A \otimes_{\mathbb{S}} A^{op}} A$.

By the work of NS, the above spectrum is in fact a cyclotomic spectrum. First, we note there is an S^1 -action on $\text{THH}(A)$, derived by following diagram.

$$N(\Lambda^{op}) \xrightarrow{V^\circ} N(\text{Ass}_{\text{act}}^\otimes) \xrightarrow{A^\otimes} \text{Sp}_{\text{act}}^\otimes \xrightarrow{\otimes} \text{Sp}.$$

Definition 4.2. A cyclotomic spectrum is pair $(X, \prod \varphi_p)$, where $X \in \text{Fun}(BS^1, \text{Sp})$, and φ_p is an S^1 -equivariant map

$$X \rightarrow X^{tC_p}.$$

Here, the $S^1 \cong S^1/C_p$ -action of X^{tC_p} is the residue action.

Remark 4.3. Let G be a compact Lie group, and $X \in \text{Fun}(BG, \text{Sp}) = \text{Sp}^{BG}$. Then the homotopy fixed point of X is defined to be $X^{hG} := \lim_{BG} X$, and the homotopy orbit of X is defined to be $X_{hG} := \text{colim}_{BG} X$. Let $f : BG \rightarrow *$ be the trivial map. Then $f_! = (-)_{hG}, f_* = (-)^{hG}, f^* = \text{triv}, f_*$ is the right adjoint of $f^*, f_!$ is the left adjoint of f^* . If moreover, G is finite, then there is a natural transformation $Nm : f_! \rightarrow f_*$. Then the Tate construction $-^{tG}$ is defined to be the cofiber of the norm map, i.e. for any $X \in \text{Fun}(BG, \text{Sp})$, $X^{tG} := \text{cofib}(Nm : f_! X \rightarrow f_* X)$. In general, if G is a topological group, then there is a norm map

$$Nm : f_!((\Sigma_+^\infty G)^{hG} \otimes X) \rightarrow f_* X.$$

If $G = S^1$, then $D_G \simeq S^1$. And the Tate construction can be also defined as a cofiber.

Definition 4.4. The topological cyclic homology TC can be defined as a fiber

$$\text{TC}(X) := \text{Fib}(X^{hS^1} \xrightarrow{\text{can} - \prod \varphi_p} \prod (X^{tC_p})^{hS^1}).$$

If X is bounded below, then $(X^{tC_p})^{hS^1} \rightarrow X^{tS^1}$ is an equivalence.

Construction 4.5. For an cyclotomic spectrum X , we define the fixed point spectrum as follows

$$\text{TR}^j(X, p) := X \times_{X^{tC_p}} X^{hC_p} \times_{X^{tC_{p^2}}} X^{hC_{p^2}} \times \dots \times_{X^{tC_{p^j}}} X^{hC_{p^j}}.$$

We define $\text{TR}(X, p) := \lim_j \text{TR}^j(X)$, there is a map $F : \text{TR}(X, p) \rightarrow \text{TR}(X, p)$. And $\text{TC}(X, p) \simeq \text{Fib}(1 - F : \text{TR}(X, p) \rightarrow \text{TR}(X, p))$.

Theorem 4.6. *Let A be an algebra, then the Dennis trace factor through TC, i.e. we have a map*

$$K(A) \rightarrow \text{TC}(A) \rightarrow \text{TC}^-(A) \rightarrow \text{THH}(A).$$

The trace map is called cyclotomic trace map, which is functorial.

The following theorem makes K-theory more computable.

Theorem 4.7 (Dundas–Goodwillie–McCarthy). *If $f : R \rightarrow S$ is a map of connective $\mathbb{E}_1$ -rings, and the map $f_0 : \pi_0 R \rightarrow \pi_0 S$ is a surjection with nilpotent kernel. Then the following diagram is a pull back square.*

$$\begin{array}{ccc} K(R) & \longrightarrow & TC(R) \\ \downarrow & & \downarrow \\ K(S) & \longrightarrow & TC(S) \end{array}$$

Theorem 4.8. *The following map is an equivalence*

$$K(R)_p^\wedge \rightarrow \tau_{\geq 0} TC(R)_p^\wedge$$

for R be a commutative ring, (R, pR) be a henselian pair such that the Frobenius $\varphi : R/p \rightarrow R/p$ is a surjection.

For instance, one could use Theorem 4.7 to compute $K(\mathbb{F}_p[x]/x^e)$. One could use Theorem 4.8 to compute $K(\mathcal{O}_K, \mathbb{Z}/p)$, where K is a finite extension of $\mathbb{Q}_p$, and $\mathcal{O}$ is the valuation ring. The computation of TC is also not easy, but it's more computable. It can be well understood thought the motivic filtration. We first review a the computation for $THH(\mathbb{F}_p)$.

Example 4.9. 1. First, we recall that the multiplicative property of THH . For R_i an $\mathbb{E}_\infty$ - k_i -algebra, where $i = 1, 2, 3$. And we have $\mathbb{E}_\infty$ -ring maps

$$R_1 \rightarrow R_2 \leftarrow R_3.$$

$$k_1 \rightarrow k_2 \leftarrow k_3.$$

We have equivalence

$$THH(R_1/k_1) \otimes_{THH(R_2/k_2)} THH(R_3/k_3) \simeq THH(R_1 \otimes_{R_2} R_3/k_1 \otimes_{k_2} k_3).$$

2. First we compute the $\mathbb{F}_p$ -homology of $THH(\mathbb{F}_p)$. By the Boksedt spectral sequence, i.e. the Tor spectral sequence for

$$\mathbb{F}_p \otimes THH(\mathbb{F}_p) \simeq THH(\mathbb{F}_p \otimes \mathbb{F}_p/\mathbb{F}_p)$$

has the formula as follows

$$A \otimes_{A \otimes_{\mathbb{F}_p} A} A,$$

where $A = \mathbb{F}_p \otimes \mathbb{F}_p$. We consider $p = 2$, the homotopy groups of A is $\pi_*(A) \simeq \mathbb{F}_2[\overline{\xi_1}, \overline{\xi_2}, \dots]$. So, the E_2 -page of the Tor spectral sequence is given by

$$\Lambda_{A_*}(d\overline{\xi_1}, d\overline{\xi_2}, \dots).$$

3. It can be showed that there is no nonzero differentials for $d\overline{\xi_i}$. We denote the class $x \in \pi_*(\mathbb{F}_p \otimes THH(\mathbb{F}_p))$ to be the lift of $d\overline{\xi_1}$.
4. Using Dyer–Lashof operation on $\pi_*(\mathbb{F}_p \otimes THH(\mathbb{F}_p))$, one shows that x^{2^r} is detected by $d\overline{\xi_{r+1}}$. Therefore, we conclude that $\pi_*(\mathbb{F}_p \otimes THH(\mathbb{F}_p)) \simeq A_*[x]$.
5. Using Adams spectral sequence, one gets that $THH_*(\mathbb{F}_p) \simeq \mathbb{F}_p[x]$.

References

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